BANKHAUL · RISK REGISTER AND HAZARD ANALYSIS

BankHaul risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	H6 controls as decided by Amish (O-2 board, O-7 lowering rope), 2026-10-25
0.1	2026-10-25	Amish Chadha	First risk register, preliminary hazard analysis (stage P1) and first-pass design FMEA (stage P3) for option A (kit 1.8 upgrade)

BankHaul risk register and hazard analysis

The top project risk is that no co-design partner and trial site are found, so the soil, the nets and the users' pull all stay assumptions. The top hazard is a crocodile attack on someone at the water's edge, above all while the far stake is placed; the station reduces time in the water but cannot protect anyone who is at the edge. The highest design FMEA scores are a weak link replaced with stronger cord (RPN 60) and the far stake leaning over in a bed softer than assumed (RPN 48). Method: STANDARDS section 23 and ISO 12100. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (unlikely to be caught).

> **Safety:** BankHaul is a rope system worked beside crocodile water. It does not protect anyone from a crocodile that leaves the water. Every control below assumes the local wildlife authority's advice is followed.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner or trial site	Candidate not yet approached (BKH-DDR-001, item 10)	Requirements, soils and nets stay desk assumptions; no TRL 4	3	5	Amish	Approach the Lake Kariba research group and an inshore cooperative before TRL 4
P2	Fishing rules do not allow a fixed set line	Gear or zoning rules (BKH-REG-001 G1)	Stations cannot be sited where attacks happen	2	5	Partner	Ask the fisheries authority before siting (PR-016)
P3	Fishers do not adopt the loop	Catch, habit, shared stations (STEM S26)	No safety benefit	3	4	Partner	Co-design with fishers; trial both shared and household stations
P4	Local-materials station costs more than USD 80	Rated rigging and rope prices (PR-008)	The station is out of reach for households	3	3	Partner	Cost it locally; share a bank post between loops (USD 269 a loop)
P5	Parts stolen	Steel and rope have resale value	Station out of use	3	3	Association	Remove the loop when not fishing; mark parts; community ownership
P6	Bed softer than 5 kPa at the site	Silt and organic mud (STEM S15)	Far stake cannot hold the design load	3	4	Partner	Hand vane from the boat before siting (BKH-DEC-001 item 7)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Crocodile attack while placing the far stake	Working on or near the water from a boat or a dry bed	Death or serious injury	5	2	Boat on two anchors or a dry bed at low water; two people and a lookout; wildlife authority's advice; never wade (BKH-BLD-001 step 6)	3
H2	Crocodile attack at the bank	Fisher near the edge at night or dawn	Death or serious injury	5	2	Working spot 5 m back (PR-011); hauling from the table; the station never asks anyone to the edge	3
H3	Fisher pulled toward the water	A crocodile or snag takes the net	Dragged in	5	2	Weak links release at 500 to 700 N (PR-010); never tie the loop or net to the body; let go	2
H4	Rope or stay snap-back	Overload, cut or worn rope, failed splice	Struck by the rope or a shackle	4	2	Factors of 9 (stays) and 11 (loop); nobody in the bight of the loop or in line with a loaded stay; inspect before each use	2
H5	Hand caught in a block or cleat	Hand near the sheave while hauling	Crushed or burnt fingers	3	2	Haul by the net's near end, not the loop; hands at least 0.5 m from the blocks	1
H6	Back strain	200 N peak pull; 33 kg stake	Injury	3	3	Peak only at the bank edge, over the smooth bank-edge board (O-2, decided 2026-10-25); the stake let down on a lowering rope by two people, not held at arm's length (O-7, decided 2026-10-25)	2
H7	Drowning	Working from a boat; falling in during installation	Death	5	2	Life jackets; two people; installation in daylight (BKH-REG-001 G12)	2
H8	Boat, swimmer or floating debris caught on the loop	A floating line across the water; the weak links do not limit a load on the loop itself	Capsize, entanglement; the stake or bank station loaded beyond design	4	2	Mark the loop and stake; take the loop out when not fishing and before floods; inspect after any hard pull	2
H10	Someone goes to the water to	The weak link released, or the	Exposure to crocodiles while wading or	5	3	Recover only from a boat with two people and a lookout, by daylight;	3

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
	recover a released net or free a fouled far block	loop jammed at the far block	reaching from a boat			otherwise abandon the net; never wade (BKH-BLD-001 safety stops)	
H11	Crocodile or hippo strikes the floating loop or the stake	Animals in the water at the station	Loop or stake overloaded; a person near the bank pulled	4	2	Nobody holds the loop while it is loaded by something unseen; let go; inspect the stake and anchors afterwards	2
H9	Welding burns and fumes	Building the post, stake and collar	Burns, eye injury	3	3	Competent welder, PPE, ventilation	1

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Weak link (BOM 7)	Limit the pull of a seized net	Replaced with stronger cord	A seized net can pull the station or the fisher	Nuisance breaks; wrong cord	5	3	4	60	Paint the link; batch test (BKH-BLD-001 3.7); spare links in the kit
F2	Far stake (BOM 2)	Hold the far block	Leans over or pulls out	Loop slackens; net lost	Bed softer than 5 kPa; the top pin hole used in a very soft bed (D-S7); scour round the stake; a firm layer stops driving short of 3.5 m	4	3	4	48	Hand vane and probe before driving (BKH-DEC-001 items 7, 12); a stake stopped short of 3.5 m gets a new check before use; check the lean each season
F3	Screw anchors (BOM 11)	Hold the stays	Creep under sustained load	Post leans toward the water	Factor near 2 in soft soil	3	3	3	27	Pull test at TRL 4 (PR-004); re-tension the turnbuckles; inspect each week
F4	Loop rope (BOM 6)	Carry the net	UV loss and abrasion	Rope breaks under load	Sun over a season (STEM S19)	4	2	3	24	Inspect before each use; replace after a season or at the first fuzzing
F5	Block bushes (BOM 8)	Let the sheave turn	Wear in silty water	Sheave wobbles, rope rubs a cheek	Silt (STEM S22)	3	3	3	27	Record wear in the season pilot; spare blocks
F6	Stop pin (BOM 14)	Hold the collar	Nut works loose, pin lost	Collar slides down the stake	Vibration, corrosion	3	2	3	18	Nyloc nut; spare pin; inspect from the boat each season
F7	Stays and turnbuckles (BOM 5, 12)	Hold the post	Turnbuckle unwinds	Post leans	Vibration of the loop	3	2	2	12	Wire the turnbuckles after tensioning
F8	Swivel rings (BOM 9)	Join the halves; act as stops	Ring reaches a block	Loop jams	Net longer than the span less 2 m	2	2	2	8	Net length rule in the build plan (PR-012)

Sources

- Matanzima, Marowa and Nhiwatiwa, *Oryx* (attack activity and timing), as cited in BKH-PRB-001.
- Northern Territory Government, Be Crocwise, fishing and boating advice (checked 2026-10-25).
- BKH-CAL-001 for the loads, factors and masses quoted.